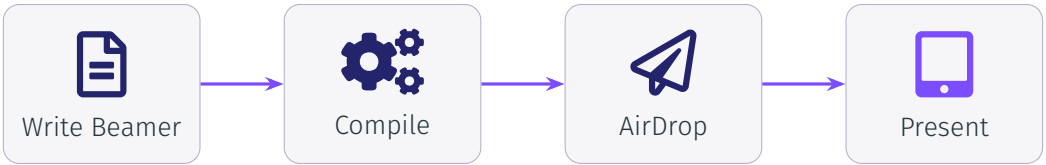
PenCast

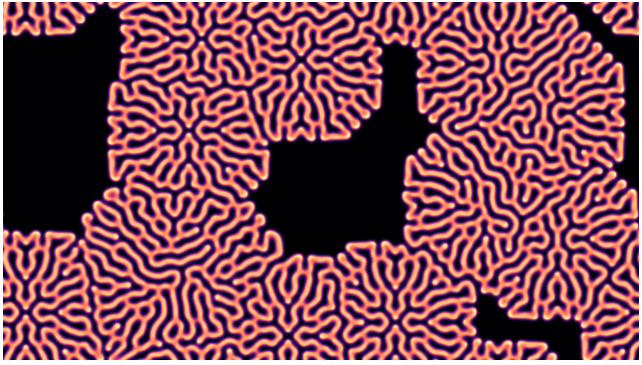
Beamer PDFs on iPad, with video, animation and ink

One PDF. No conversion.



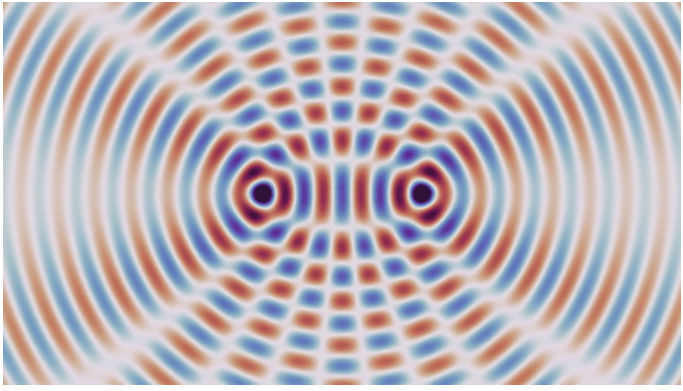
Videos, animations, links and overlays come straight from your PDF:
no PowerPoint step, no sidecar files, no special macros.

Embedded videos play in place



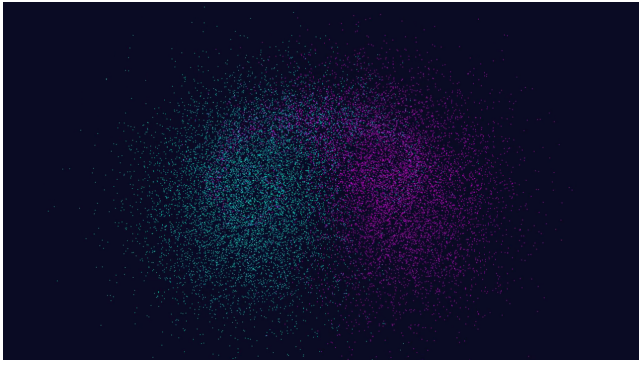
- MP4s embedded with `media9` or a standard PDF Screen annotation
- Autoplay and loop settings come from the PDF
- iPad and projector play the same video, in sync

Write right over a playing video



👉 *Circle the bright fringes where the two waves add up. Ink sits above the video on both screens.*

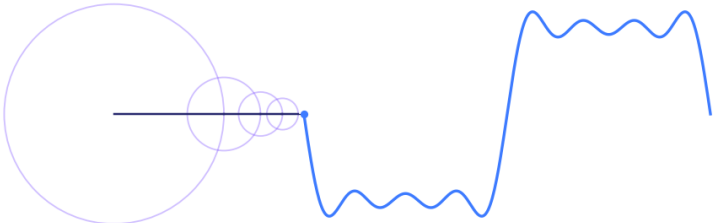
Tap a video to control it



- Tap: play / pause
- Drag to scrub through it
- Controls stay on the iPad, never on the projector

An unequal counter-rotating vortex pair stirring two clouds of tracers.

`animate` sequences, frame-accurate



A square wave from its first four harmonics, $\frac{4}{\pi}\sum_{k\,\text{odd}}\frac{\sin kt}{k}$.

Frame rate and looping are read from the PDF. Tap the animation to pause it.

Your notes survive every \pause

$$I = \int_{-\infty}^{\infty} e^{-x^2}\,dx$$

Annotate the first step:

your ink stays as each line appears.

Write on step one, then advance.

Your notes survive every \pause

$$I = \int_{-\infty}^{\infty} e^{-x^2}\,dx$$

$$I^2 = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)}\,dx\,dy$$

Annotate the first step:

your ink stays as each line appears.

Write on step one, then advance.

Your notes survive every \pause

$$
\begin{aligned}
I &= \int_{-\infty}^{\infty} e^{-x^2}\,dx \\
I^2 &= \iint_{\mathbb{R}^2} e^{-(x^2+y^2)}\,dx\,dy \\
&= \int_0^{2\pi}\int_0^{\infty} e^{-r^2}\,r\,dr\,d\theta
\end{aligned}
$$

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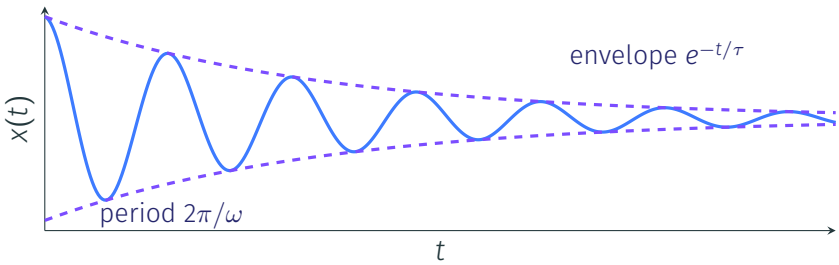
$$
\begin{aligned}
I &= \int_{-\infty}^{\infty} e^{-x^2}\,dx \\
I^2 &= \iint_{\mathbb{R}^2} e^{-(x^2+y^2)}\,dx\,dy \\
&= \int_0^{2\pi}\int_0^{\infty} e^{-r^2}\,r\,dr\,d\theta \\
&= \pi \quad\Longrightarrow\quad I = \sqrt{\pi}
\end{aligned}
$$

Annotate the first step:

your ink stays as each line appears.

Write on step one, then advance.

Point with the laser



👉 *Pick the laser: a glowing trail while you drag, gone when you lift. Or just a dot.*

Zoom in to write neatly

17	48	79	21	52	83	25	56	87	29
34	65	96	38	69	11	42	73	15	46
51	82	24	55	86	28	59	90	32	63
68	10	41	72	14	45	76	18	49	80
85	27	58	89	31	62	93	35	66	97
13	44	75	17	48	79	21	52	83	25
30	61	92	34	65	96	38	69	11	42
47	78	20	51	82	24	9.07	86	28	59
64	95	37	68	10	41	72	14	45	76
81	23	54	85	27	58	89	31	62	93
98	40	71	13	44	75	17	48	79	21
26	57	88	30	61	92	34	65	96	38

- Pinch to zoom; one finger moves the page while you write
- The projector can keep the whole slide, or follow your zoom
- Stray taps can't change slides while zoomed

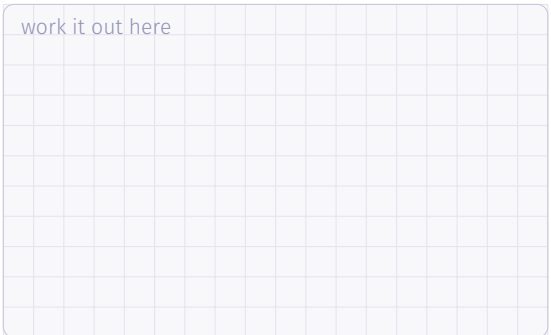
One number in the table is not a whole number. Zoom in and circle it.

Lecture notes with a real fountain pen

Example. Find the eigenvalues of

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

- Round nib that follows Pencil pressure
- Tune how hard you press for thick and thin
- Lasso to move what you wrote



Links and navigation just work

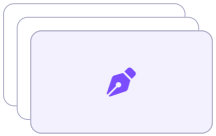
▸ Jump to the appendix

Hyperref links, Beamer's navigation symbols, a Bluetooth clicker, arrow keys, swipes, or the slide grid: whatever you already use.

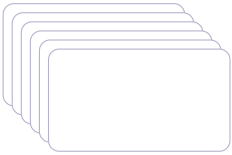
Made for the room

- Projector shows slide, video and ink; no toolbar
- HDMI or AirPlay; replug mid-talk without restarting
- Black, white or freeze the projector
- Talk timer
- Beamer speaker notes, on the iPad only
- The same app on the Mac

Take your notes with you



Notes
one page per slide, final step, with your ink



Every page
every overlay step, as presented

Tap to save, share (AirDrop, Mail) or print, and send it to your students.

PenCast

Write Beamer. Present with ink.

Appendix: why $\int e^{-x^2}\,dx = \sqrt{\pi}$ needs polar coordinates

e^{-x^2} has no elementary antiderivative, so the trick is to square the integral and switch to polar coordinates, where the extra factor r makes $e^{-r^2} r$ integrable:

$$\int_0^\infty e^{-r^2} r\,dr = \tfrac{1}{2}.$$

◂ Back